

Pearson Edexcel Mark Scheme

Predicted Paper — For revision purposes only

Level 3 GCE Economics A (9EC0) Paper 1: Markets and Business Behaviour — 9EC0/01 — Hard Total: 100 marks

General Marking Guidance

- All candidates must receive the same treatment. Mark schemes should be applied positively, rewarding candidates for what they have shown they can do.
- All marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.
- Accept any equivalent wording and alternative valid approaches not explicitly listed in the indicative content.

Assessment Objectives

AO1 Demonstrate knowledge of terms, concepts, theories and models.

AO2 Apply knowledge and understanding to various economic contexts.

AO3 Analyse issues within economics, using chains of reasoning.

AO4 Evaluate economic arguments and use evidence to support informed judgements.

Marking grids are organised as KAA (AO1+AO2+AO3) and Evaluation (AO4).

Level Descriptors

8-mark "Examine" (KAA = 5, Evaluation = 3)

Level	KAA marks	Descriptor
L1	1–2	Isolated element of knowledge; limited application.
L2	3–4	Sound knowledge applied with some development; at least one chain of reasoning.
L3	5	Accurate detailed knowledge fully applied; developed chains.

10-mark "Assess" (KAA = 6, Evaluation = 4)

Level	KAA marks	Descriptor
L1	1–2	Limited knowledge; weak application.
L2	3–4	Sound theory applied to context.
L3	5–6	Accurate detailed theory, well applied; developed chains.

12-mark "Discuss with diagram" (KAA = 8, Evaluation = 4)

Level	KAA marks	Descriptor
L1	1–2	Limited knowledge; weak or missing diagram.
L2	3–5	Sound knowledge; reasonable diagram; some application.

Level	KAA marks	Descriptor
L3	6–8	Accurate knowledge; correctly labelled diagram; developed chains.

15-mark "Discuss" (KAA = 9, Evaluation = 6)

Level	KAA marks	Descriptor
L1	1–3	Limited knowledge; weak application.
L2	4–6	Sound knowledge with application; some development.
L3	7–9	Accurate detailed knowledge; developed chains; relevant diagrams.

25-mark Essay "Evaluate" (KAA = 15, Evaluation = 10)

Level	KAA marks	Descriptor
L1	1–5	Limited knowledge; basic theory.
L2	6–9	Sound theory applied to context.
L3	10–12	Accurate detailed theory, well applied; relevant diagrams.
L4	13–15	Comprehensive, sophisticated theory; diagrams used analytically.

25-mark Essay evaluation grid (AO4)

Level	Eval marks	Descriptor
L1	1–3	Assertion without support.
L2	4–6	Supported evaluation; some judgement.
L3	7–8	Developed evaluation; reasoned judgement.
L4	9–10	Sophisticated evaluation with prioritised judgement.

Section A — Questions 1 to 5

Question 1 (5 marks) — UK ETS carbon permits

Q	Indicative content	Marks
1(a)	Method: $PES = \% \Delta Qs \div \% \Delta P$ (1). $\% \Delta Qs = -12\%$ (cap reduced permits issued). $\% \Delta P = (£72 - £45) \div £45 \times 100 = +60\%$ (1). $PES = -12 \div 60 = -0.20$ (accept 0.2 in absolute value; accept answers in the range -0.18 to -0.22 with valid working) (2). Accept alternative valid interpretation: because the quantity change was driven by a regulator-set cap rather than a movement along a supply curve, candidates may argue that supply is effectively price-inelastic or vertical over this range. Award full credit for any internally-consistent calculation of the ratio of $\% \Delta Q$ to $\% \Delta P$ with supporting working, even if the sign interpretation differs.	4
1(b)	Answer: B — invest in lower-emission production technology. A higher permit price raises the marginal cost of polluting; firms have a direct incentive to substitute toward cleaner production methods whose return now exceeds the cost of permits avoided. Distractors: A incentive runs the other way — higher permit costs discourage expansion of output that generates emissions; C wages for low-skilled workers are not directly linked to permit pricing; D firms would prefer subsidies, but the permit-price signal itself does not generate one — it incentivises internal abatement.	1

Question 2 (5 marks) — Pension auto-enrolment and bounded rationality

Q	Indicative content	Marks
2(a)	Bounded rationality: workers have limited cognitive capacity, time and information; decisions about long-term savings involve complex uncertainty (1). Choice architecture: default settings anchor behaviour; in the opt-in condition, inertia and status-quo bias keep participation low at 38%; in the opt-out condition, the same inertia produces the opposite outcome and 91% remain enrolled (1). Evidence that workers do not optimise over lifetime utility independently of the framing — consistent with bounded rationality (1). Present bias may also contribute: workers systematically under-weight future benefits of pension saving (1).	4
2(b)	Answer: B — status-quo bias effect. The same set of workers reaches opposite outcomes purely because the default option has changed; this is the defining pattern of status-quo bias, where individuals disproportionately stick with whatever option has been set as the default. Distractors: A positive externalities relate to third-party benefits in production or consumption, not to decision-framing; C moral hazard describes post-contract behavioural change when actions are unobservable, which is not what this experiment captures; D Giffen paradox concerns an upward-sloping demand curve for a specific category of inferior good and is not relevant here.	1

Question 3 (5 marks) — Supermarket petrol game theory

Q	Indicative content	Marks
3(a)	Nash equilibrium: (Cut, Cut) with payoffs (4, 4). At this strategy pair, neither firm can improve its payoff unilaterally. Working: if A: Cut, B: Maintain → B gets 2, better to Cut (11 → 4? No — 4 > 2, so Cut). If A: Maintain, B: Cut → B gets 11 > 8 → Cut. So B's dominant strategy is Cut. Symmetric for A. Unique NE: (Cut, Cut). Award 1 mark for identifying dominant strategies / best responses; 1 mark for correct NE identified with payoffs.	2
3(b)(i)	Combined weekly profit at NE = 4 + 4 = £8 million.	1
3(b)(ii)	Combined profit if both Maintain = 8 + 8 = £16 million. Illustrates prisoner's-dilemma structure — cooperation Pareto-dominates the NE.	1
3(c)	Answer: C — interactions are frequent, credible punishments exist, and firms value future profits highly. Folk theorem of repeated games: collusion sustained when discount factor sufficient to make defection's short-term gains smaller than long-term losses from trigger strategies.	1

Question 4 (5 marks) — Monopsony and the minimum wage

Q	Indicative content	Marks
4(a)	Diagram: upward-sloping labour supply ($S_L = ACL$); MCL curve steeper above ACL because monopsonist pays all workers the higher wage to hire one more. Labour demand $D_L = MRPL$ (1). Monopsony equilibrium: firm hires where $MCL = MRPL$ at Q_m; pays wage W_m read off the supply curve, below $W_{competitive}$ (1). Impose minimum wage W_{min} where $W_m < W_{min} \leq$ competitive wage: horizontal MCL up to the labour supply curve at W_{min}. Firm now hires up to where $W_{min} = MRPL$, giving $Q_{min} > Q_m$ (1). Therefore both the wage and employment rise — opposite to the competitive case. Explain intuition: minimum wage removes the monopsonist's ability to set wage below MRPL (1).	4
4(b)	Answer: B — a rise in wages and a fall in employment, creating excess labour supply. Standard competitive-market outcome: binding minimum wage raises quantity supplied above quantity demanded.	1

Question 5 (5 marks) — NLW and grocery retail wage bill

Q	Indicative content	Marks
5(a)	Answer: C — relatively elastic (greater than one in absolute value). When demand for labour is elastic, a wage rise produces a proportionately larger fall in employment.	1
5(b)	Acceptable reasons (any ONE developed, 2 marks for definition + chain): • Few close substitutes for low-skilled shop-floor labour in the short run	2

Q	Indicative content	Marks
	(checkout staff, shelf-stackers) — automation capital-intensive, limited (1+1). • Labour costs a small share of total costs for large chains → wage rise causes only modest increase in unit costs → limited output/employment effects (1+1). • Derived-demand: demand for groceries price-inelastic, so retailers pass cost increases through to prices without major quantity effects (1+1).	
5(c)	Wage bill per worker per week old: $£11.44 \times 30 = £343.20$. New: $£12.21 \times 30 = £366.30$. Rise per worker: $£366.30 - £343.20 = £23.10$ per week. For 400 workers: $400 \times £23.10 = £9,240$ per week. Alternative method: rise in hourly rate = $£0.77$; $\times 30$ hours $\times 400$ workers = $£9,240$. 1 mark method; 1 mark correct numerical answer with unit.	2

Section B — Question 6: UK retail banking

6(a) — Four-firm concentration ratio and consequences (5 marks)

Indicative content (KAA 3 / Eval 2):

Top four = Lloyds 25.4 + NatWest 17.2 + Barclays 14.1 + HSBC 10.8 = 67.5% — highly concentrated tight oligopoly (2). Consequences may include: reduced price competition (e.g. low interest on current-account balances), weakened incentive to invest in customer service, possible implicit price coordination (2).

Evaluation (1): challenger-bank growth from <2% (2019) to 8.6% (2024) suggests CR4 concentration overstates the lack of competitive pressure at the margin.

6(b) — Effects of Open Banking on competition (12 marks)

Diagram required: labour market or cost/revenue diagram showing how reduced barriers to entry raise number of firms/output and reduce mark-ups. A kinked-demand or oligopoly-with-reduced-MC diagram is acceptable.

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
• Open Banking enables third-party access to transaction data with consent — 11.8m users by 2024 (Fig 2). • Reduces consumer switching costs and search costs. • Lowers fixed-cost barriers to entry for fintech challengers (no need to build full banking stack). • Enables product innovation: budgeting tools, affordability checks, FX services — raises non-price competition. • Challenger share rose from <2% (2019) to 8.6% (2024) — Fig 1 and Extract B. • Diagram: new entrants shift industry supply curve right → lower equilibrium price/commission.	• Growth is real but concentrated at the margin — Big 4 still hold ~67.5% of PCAs (Fig 1). • Network effects and brand trust favour incumbents (particularly older demographics). • Take-up of Open Banking required years to reach scale — not a quick fix. • Consumer inertia and status-quo bias (Extract A) limit actual switching even when costs are zero. • Judgement: Open Banking has increased competition at the margin, particularly in digital-native and fee-free segments. But structural dominance of incumbents in deposits, mortgages, and cross-selling remains largely intact. Meaningful competition requires additional structural measures.

6(c) — Two measures other than Open Banking to strengthen competition (8 marks)

Indicative content (KAA 5 / Eval 3). Accept any two, fully developed:

1. Structural remedies (divestiture / forced break-up) — CMA can order incumbents to sell branch networks or customer portfolios. Chain: directly reduces concentration; precedent exists (Lloyds forced to divest TSB in 2013) (2–3).
2. Account number portability (like mobile number portability) — removes friction entirely. Chain: eliminates technical switching costs, making switching as easy as changing mobile provider (2–3).
3. Cap on overdraft fees or other behavioural-pricing products — reduces rent extraction (2).
4. Promoting mutual/co-operative banks and credit unions via tax reliefs and simplified regulation (2).

Evaluation (3): risks — structural remedies may reduce economies of scale; portability extremely expensive to build; caps may cause product withdrawal. Best option depends on empirical elasticities and cost-benefit analysis. Judgement: account portability likely the highest-impact intervention, combining direct removal of friction with preservation of scale efficiencies.

6(d) — Behavioural biases vs. structural market power (10 marks)

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
• Behavioural case (Extract A): status-quo bias, hassle costs, limited engagement keep switching <2%. • If inertia is the primary constraint, structural remedies may not raise competition — consumers won't use the alternatives. • Nudges (default switching, simplified comparison) may be more effective than structural reform. • Evidence: Open Banking has raised tools available but not dramatically raised switching rates. • CASS (Current Account Switch Service) makes switching technically easy but has not shifted behaviour.	• Structural case: concentration (CR4 = 67.5%) and brand incumbency reflect genuine market power — not purely behavioural. • Scale economies, cross-selling and branch networks create real barriers. • Removing biases doesn't help if there's no meaningful choice — need both sides of the reform. • Recent challenger-bank growth shows structural change IS driving shift (8.6% share in 2024). • Judgement: behavioural biases amplify market power but do not cause it. Primary obstacle is structural — without credible challengers, behavioural nudges have nothing to nudge towards. Competition policy needs both pillars: structural reform to create alternatives AND behavioural reform to prompt engagement. Neither alone is sufficient.

6(e) — Impact of Starling/Monzo merger on consumers (15 marks)

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
• Merger raises combined challenger share to 6.1% — becomes the largest digital-only bank but still below each Big-4 incumbent. • Horizontal integration reduces direct competition between the two leading digital-natives. • Potential increase in fees/decreased non-price competition in fee-free segment. • Combined scale may enable investment in machine-learning credit assessment, better risk management (Extract C). • Could create a credible fifth force to rival the Big 4 — raising overall competition at sector level. • Possible reduction in choice among digital-native propositions (Extract C).	• Effect on consumers depends on CMA remedies: divestiture of overlapping products may mitigate concern. • Competition between Big 4 and challengers may matter more than between Starling and Monzo. • Dynamic efficiency gains likely — both firms are unprofitable individually; merger may reach sustainable scale. • Risk of duopoly in digital-native segment — remaining digital challengers (Revolut, Chase UK) may fill the gap. • Long-term consumer surplus likely improved if merger accelerates displacement of incumbents. • Judgement: merger creates short-run reduction in choice among digital challengers but long-run gain if combined entity can meaningfully contest the Big 4.

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
	CMA should clear subject to conduct remedies (e.g. keep separate brands for 3–5 years, portability commitments). On balance, likely net positive for consumers if regulated carefully.

Section C — Question 7 OR Question 8 (25 marks)

Question 7 — Nudges vs Pigouvian taxation for health market failures (25 marks)

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
• Pigouvian framework: taxes internalise externalities by raising private marginal cost to social marginal cost. • Applied to obesity: Soft Drinks Industry Levy (2018) — £0.18 per litre on high-sugar drinks. • Nudges: choice architecture to harness behavioural biases. Default menu sizing exploits default bias; warning labels address information failures. • Diagram: tax shifts S left and reduces Q to Q_{social} (allocative efficiency restored). • Evidence on SDIL: reformulation successful; consumption of sugar from soft drinks fell ~30%. • Behavioural insights: traffic-light labelling, portion-size defaults, restaurant menu calorie labelling.	• Nudges advantages: low cost, preserve choice, address information failures directly, no regressive impact. • Pigouvian advantages: raise revenue, provide continuous incentive, transmit through price system. • Nudges disadvantages: modest effect sizes, may not overcome deeply addictive behaviours (alcohol, tobacco). • Pigouvian disadvantages: regressive (falls on lower-income consumers); may cause cross-border substitution; risk of illicit markets. • Asymmetry of the externality: health effects individualised, whereas Pigouvian framework assumes clear third-party harm. • Judgement: nudges and taxation are complementary, not substitutes. Taxation better where external costs are large and clear (sugar, tobacco); nudges better where information failures and present bias are primary causes. Best policy mix combines both — SDIL + menu labelling + default portion sizing. Pure reliance on either alone is suboptimal.

Question 8 — Algorithmic pricing and tacit collusion (25 marks)

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
• Concept: algorithmic pricing — automated adjustment of prices based on competitor data, demand signals, and machine learning. • Theoretical basis: Q-learning and reinforcement-learning algorithms converge on supra-competitive prices in repeated games without explicit agreement. • Example industries: online retail (Amazon marketplace), petrol forecourts, airline tickets, dynamic-pricing ride-hailing. • Competition law implication: traditional cartel law requires proof of agreement; algorithms make this difficult. • Possible regulatory responses: per-se	• Counter: algorithms also enhance competition via faster price adjustments, better price discovery, lower search costs. • Empirical evidence mixed: Assad et al. (2023) find algorithmic tacit collusion in German petrol forecourts; other studies find purely competitive outcomes. • Prohibition may chill innovation and raise compliance costs; risks extensive false positives. • Alternative: outcome-focused enforcement (investigate markets where margins rise without explicit agreement). • Jurisdictional issue: algorithms run across borders; unilateral UK ban may be ineffective.

KAA (AO1 + AO2 + AO3)	Evaluation (AO4)
prohibition of certain algorithm types; algorithmic transparency; periodic auditing; outcome-based tests (e.g. prices consistently above Bertrand-Nash predictions).	• Judgement: absolute prohibition is too broad — it would capture pro-competitive algorithms alongside collusive ones. Smarter regulation: targeted scrutiny in concentrated markets where collusion is theoretically most likely (≤ 4 competitors, transparent pricing); sandbox testing of algorithms before deployment; an outcome-based market investigation tool. The claim that algorithmic pricing 'inevitably' leads to collusion is an overstatement.

TOTAL FOR PAPER = 100 MARKS